JavaScript RegExp Assertions

[Back to JS page](#)

Table of Contents

RegExp Assertions

Assertions are conditions that must be true at a position in the string:

Assertion	Description
`^`	Matches the beginning of the string
`$`	Matches the end of the string
`\b`	Matches at a word boundary
`x(?=y)`	Lookahead: x only if followed by y
`x(?!y)`	Negative lookahead: x only if NOT followed by y
`(?<=y)x`	Lookbehind: x only if preceded by y
`(?<!y)x`	Negative lookbehind: x only if NOT preceded by y

RegExp ^ Metacharacter

`^` asserts the position at the start of the string:

```html
<!DOCTYPE html>
<html>
<body>

<h2>JavaScript RegExp Assertions</h2>
<p>From w3schools.com, Experiment by Teeratus_R</p>

<h4>RegExp ^ Metacharacter</h4>
<p id="demo"></p>

<script>
const text = "Hello World, Hello Again";
const matches = text.match(/^Hello/g);

document.getElementById("demo").innerHTML =
  "Text: " + text + "<br><br>" +
  "match(/^Hello/g): " + matches + "<br>" +
  "^ matches 'Hello' only at the start";
</script>

</body>
</html>
```

Example 1

Result [View Example]

RegExp $ Metacharacter

`$` asserts the position at the end of the string:

```
<!DOCTYPE html>
<html>
<body>

<h2>JavaScript RegExp Assertions</h2>
<p>From w3schools.com, Experiment by Teeratus_R</p>

<h4>RegExp $ Metacharacter</h4>
<p id="demo"></p>

<script>
const text = "Hello World, Hello";
const matches = text.match(/Hello$/g);

document.getElementById("demo").innerHTML =
  "Text: " + text + "<br><br>" +
  "match(/Hello$/g): " + matches + "<br>" +
  "$ matches 'Hello' only at the end";
</script>

</body>
</html>
```

Example 2

Result [View Example]

The \b Metacharacter

`\b` asserts a word boundary - where a word starts or ends:

```
<!DOCTYPE html>
<html>
<body>

<h2>JavaScript RegExp Assertions</h2>
<p>From w3schools.com, Experiment by Teeratus_R</p>

<h4>The \b Metacharacter</h4>
<p id="demo"></p>

<script>
const text = "car cart scar";
const matches = text.match(/\bcar\b/g);

document.getElementById("demo").innerHTML =
  "Text: " + text + "<br><br>" +
  "match(/\\bcar\\b/g): " + matches + "<br>" +
  "Finds 'car' as a whole word only";
</script>

</body>
</html>
```

Example 3

Result View Example

RegExp Lookahead x(?=y)

Lookahead matches `x` only if it is followed by `y`:

```
<!DOCTYPE html>
<html>
<body>

<h2>JavaScript RegExp Assertions</h2>
<p>From w3schools.com, Experiment by Teeratus_R</p>

<h4>RegExp Lookahead x(?=y)</h4>
<p id="demo"></p>

<script>
const text = "apple banana applepie";
const matches = text.match(/apple(?=pie)/g);

document.getElementById("demo").innerHTML =
  "Text: " + text + "<br><br>" +
  "match(/apple(?=pie)/g): " + matches + "<br>" +
  "Matches 'apple' only when followed by 'pie'";
</script>

</body>
</html>
```

Example 4

Result View Example

Negative Lookahead x(?!y)

Negative lookahead matches `x` only if it is NOT followed by `y`:

```
<!DOCTYPE html>
<html>
<body>

<h2>JavaScript RegExp Assertions</h2>
<p>From w3schools.com, Experiment by Teeratus_R</p>

<h4>Negative Lookahead x(?!y)</h4>
<p id="demo"></p>

<script>
const text = "apple banana applepie";
const matches = text.match(/apple(?!pie)/g);

document.getElementById("demo").innerHTML =
  "Text: " + text + "<br><br>" +
  "match(/apple(?!pie)/g): " + matches + "<br>" +
  "Matches 'apple' only when NOT followed by 'pie'";
</script>

</body>
</html>
```

Example 5

Result View Example

Document

Document in project

You can Download PDF file.

Reference

- W3Schools JavaScript RegExp Assertions